NUS School of Computing

FT5007 Capstone

Month < August 2026 > Report by Individual Team Member

Project Title: Agentic AI Chatbot for Consumer Banking

Reporting Period: August 2026

For each team member, please describe your achievements, upcoming plans (to-dos), challenges, and plans to overcome challenges in this period.

Student Name: Wu Qianzong

Achievement during the month:

- **Frontend:**
  - **Conversation experience:** refined the chat interface into a responsive banking-assistant experience with automatic scrolling, timestamps, chat history, next action options, progress, list, clarification and decline response blocks.
  - **Guided and action UI:** introduced progress views, list presentations, clarification cues, escalation confirmation, soft-escalation banner, native bank page link, and embedded service-request forms. The interface now adapts the presentation to the customer need rather than using a fixed response layout.
  - **Personalisation and usability:** added user preferences, visible node-progress wording, and feedback controls for quality assessment.
- **Backend:**
  - **Workflow and orchestration:** evolved the LangGraph workflow from the July single-journey flow into a tool-calling, multi-agent orchestration architecture. The current design resolves the true request from context, decomposes mixed turns, keeps conversational components separate, and aggregates all tool results into one coherent response.
  - **Tool Calling v2.0/v3.0:** implemented a registry-backed, extensible tool layer with subquery-owned attempts. RAG, deterministic deeplinks, protected personal-data lookup, and low/intermediate-risk service requests can be planned in parallel and reconciled at the financial-subquery level.
  - **Safety and quality controls:** implemented inbound Prompt Guard screening; triage of non-compliant, escalation, financial and non-financial components; precision substitution; and a separated evidence-groundedness and conduct-compliance audit with retry and safe fallback paths.
  - **Escalation:** implemented hard escalation confirmation with an audited case summary and a non-blocking soft-escalation.
- **RAG pipeline:**
  - **Knowledge-base pipeline:** completed and documented the FAQ staging, product-intent taxonomy, parent-child storage, placeholder/precision mapping, SQLite parent hydration and Chroma vector-index ingestion workflow for the synthesized FAQ knowledge base.
  - **Retrieval and evidence control:** implemented task-isolated retrieval, partial-evidence retention, evidence participation in ranking, and a sufficiency auditor that distinguishes sufficient, partially sufficient and insufficient evidence before information is exposed downstream.

    - **Research and observability:** created structured system logging for prompts, responses, graph events and memory activity; performed retrieval experiments comparing direct and multi-angle query representations; and recorded the resulting ETL and search-strategy decisions in maintained project documentation.
- **Memory and session handling:**
    - **Session-safe context:** implemented in-session rolling compaction with summary validation. Cross session memory in greeting by checking unresolved issues and pending service forms upon initialization.

To-dos in next month:

September will focus primarily on measurable validation and refinement. The system has reached a broad multi-tool MVP state; the next step is to use scenarios to show whether it is reliable, grounded, efficient and able to resolve realistic banking needs.

Key Implementation Tasks:

1. **Metrics harness and evaluation dataset**
    - Implement reproducible synthetic multi-turn scenarios and telemetry extraction. Report containment, first-contact resolution, triage/decomposition accuracy, evidence groundedness/conduct compliance and latency with assumptions, denominators and failure categories.
    - Use an independent judge for groundedness evaluation where the generation/judge model routing permits it; document model versions, rate-limit handling, deterministic precision-token treatment and any unjudged/error cases.
2. **RAG search-strategy intelligence and refinement**
    - Validate the current extra-effort/search-territory design and improve it only where trace evidence shows a measurable gain.
    - Research on fully LLM-automated reasoning to guide the search. (Interested to explore, cannot guarantee delivery in MVP)
3. **Complete safety, compliance and response refinement**
    - Develop Compliance v2.0 to distinguish customer informational questions from requests that ask the bank to perform a prohibited action, and make boundary responses more natural while retaining strict no-detail refusal behaviour.

Challenges Faced

1. **Natural response vs deterministic underlying control:** The deterministic schema that governs the response ensure the stability of response quality but also introduced machine response feeling, sometimes some deterministic fallback even cause response to be incoherent.
2. **User ambiguity:** User ambiguity will cause retrieval ambiguity. It is hard to draw a line between ambiguity and let the bot know the exact time to raise a clarification.
3. **Retrieval ambiguity:** The rule to determine sufficient, partly sufficient and insufficient is clear, but to reason to find what might be relevant that user might be interested in during retrieval is hard.

Plans to Overcome Challenges

1. **Natural response vs deterministic underlying control:** Decouple underlying info structure from the final response. Response is customer facing and free form.

2. **User ambiguity:** Clearly define different layer of agent's work scope and ask them to raise clarification if any ambiguity arise that will block them to perform their work in their scope. Mandatory reasoning before generation to ensure quality.
3. **Retrieval ambiguity:** Mandatory reasoning before sufficiency auditing to ensure model audit quality.

Appendix:

Project Backlog:

https://github.com/users/wuqianzong/projects/2/views/1

Version control:

Repository: https://github.com/wuqianzong/AgenticBanking

August implementation record: 100 commits (1-31 August 2026). The commit history covers workflow evolution, RAG/ETL research, memory and safety controls, user-interface refinement, deeplink/API/service-request capabilities, and Tool Calling v2.0/v3.0.

Representative August delivery trail:

Period	Representative delivery evidence
1-9 Aug	Initial RAG/data staging, parallel workflow and multi-model orchestration; semantic and deterministic safety gates; precision substitution.
12-15 Aug	Chat-history recall, rolling/long-term memory work, user journey v2.0, progress UI and validated memory summaries.
16-23 Aug	Evidence-manifest discipline, task-isolated RAG retries, response blocks, groundedness v2.0, escalation confirmation and Tool Calling v2.0.
24-27 Aug	Deeplink and personal-data capability demonstrations, service-request workflows, subquery-owned Tool Calling v3.0, soft escalation and feedback.
28-31 Aug	Dynamic RAG retry-effort research, unified clarification/decline delivery, grounding improvements, and final UI simplification.

Supporting project specifications updated or produced in August:

- End-to-End Process Specification
- RAG Specification and Search Strategy Specification
- Tool Calling v2.0 and Tool Calling v3.0
- Groundedness v2.0
- Memory Architecture v3.0
- RAG ETL Specification and ETL History and Artifact Guide
- Service Request Capability Design, Deep Link Design and Soft Escalation Design
- Metrics & Telemetry Reference Guide v3.5 (September evaluation reference; targets subject to refinement)